Type MS · Compare sorts and searches

Answer 1 of 8 · Question 3 part b · 2 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part b in this mark scheme extract

3(a)(i)	A1 A2	Award **one** mark for each: • Pass 1 correct (1) • Pass 2 correct (1) 9 11 12 19 14 18 15 17 9 11 12 14 18 15 17 19 (1) 9 11 12 14 15 17 18 19 (1) Must show ONLY the 19 moving in first pass and ONLY the 18 moving in second pass	Ignore lines with individual swaps and only award correct completed passes 9 11 12 19 14 18 15 17 9 11 12 **14** **19** 18 15 17 9 11 12 14 **18** **19** 15 17 9 11 12 14 18 **15** **19** 17 9 11 12 14 18 15 **17** **19** (1) 9 11 12 14 **15** **18** 17 19 9 11 12 14 15 **17** **18** 19 (1) Accept **accurate** right-to-left bubble sort as shown below 9 11 12 19 14 18 15 17 9 11 12 14 19 15 18 17 (1) 9 11 12 14 15 19 17 18 9 11 12 14 15 17 19 18 9 11 12 14 18 15 17 19 (1) Complete sort cannot be the 1st /only line in response – 0 marks as this will be taken as both 18 and 19 moving in first pass	**2**

Type MS · Compare sorts and searches

Answer 1 of 8 · Question 3 part b · 2 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
3(a)(ii)	A3	Award **one** mark for: • 4	If right-to-left pass completed in Q3ai, can only accept 2 swaps	1

Question	mp	Answer	Additional Guidance	Mark
3(a)(iii)	A4	Award **one** mark for: • 3	Do not accept 2 – 3rd pass required for **algorithm** to finish If right-to-left pass completed in Q3ai, can only accept 5 passes	1

Question	mp	Answer	Additional guidance	Mark
3(b)	B1	Award **one** mark for: • Binary search	Accept any other divide and conquer search algorithm (google if necessary) Accept 'binary' on its own as search in question	2
	B2	Award **one** mark for any of the following: • Divide and conquer / list halves each time(1) • Fewer searches needed (1)	Do not accept points about being 'already sorted' as this is in question	

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Type MS · Compare sorts and searches

Answer 1 of 8 · Question 3 part b · 2 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
3(c)		Award **one** mark for each of:		
	C1	(while) Loop ends **only** when 0 entered (1)	Any method e.g. `while age != 0:`	

	C2	Avoiding divide by zero error (1)	By coding 2nd selection box from flowchart – no output required
			Must come after some attempt to process age or count – award even if inside loop.
			ignore 'other' divide by zero errors introduced by candidate
			Must follow logic of flowchart e.g.:
			`if count == 0: (or count != 0)`
			…
			`else`
			…	

	C3	Average value rounded to whole number (1)	Must come after some attempt to process age or count – award even if inside loop. Must target calculated average (even if formula not correct) Any method to get int Accept casting to int *(round not in spec, but string manipulation is)* Accept integer division	**5**

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Type MS · Compare sorts and searches

Answer 1 of 8 · Question 3 part b · 2 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part b in this mark scheme extract

	C4	Correct value calculated for average and count (1)	Allow even if 0 used to initialise count in B1 (this must be only error in calculation) – set count to -1 to test	
	C5	Count and average included in a syntactically correct final print statement (1)	Award even if never runs Must include both variables (may also include string)	

Correct

```
# Q03c

# program variables
age = 1
count = -1
total = 0

# ------------------------------------------------
# Write your code below this line

while age != 0:

    age = int(input("Please enter age in years: "))

    total = total + age
    count = count + 1

if count == 0:
    print("No ages input")
else:
    average = round(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Typical

```
# Q03c

# program variables
age = 0
count = 0
total = 0

# ------------------------------------------------
# Write your code below this line
age = int(input("Please enter age in years: "))
while age != 0:

    total = total + age
    count = count + 1

    age = int(input("Please enter age in years: "))

if count == 0:
    print("No ages input")
else:
    average = int(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Type MS · Compare sorts and searches

Answer 1 of 8 · Question 3 part b · 2 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 1 of 17 · find part b in this mark scheme extract

Q3(b)	More memory/storage required (1) Because not sorted in place / not efficient / many steps / division of data set / is recursive (1)		(2)

Question	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 2 of 17 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark
Q3(a)	Award 1 mark each for: 1. File opened for input (1) 2. File opened for output (1) 3. Loop for processing data (1) 4. Check for '@' symbol (1) 5. Write invalid address to output file (1) 6. Close files (1)	Amended code should be saved as Q03aFINISHED. Do not penalise candidates who attempt more than the stated requirements.	(6)

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 3 of 17 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 4 of 17 · find part b in this mark scheme extract

Q3(c)	Award 1 mark for each section of code that: 1. Asks for input of a number and stores it in a variable with a meaningful name (1) 2. Checks number is between 1 and 50 (1) 3. Calculates square and cube of number (1) 4. Displays suitable labels e.g. 'Number', 'Square' and 'Cube' and relevant values (1) For the whole code: 5. Compiles and runs without logical or syntax errors (1) 6. Efficient in terms of computation, storage and selection of programming constructs (1)	Amended code should be saved as Q03cFINISHED. Allow follow through for mark point 4 when square and cube are incorrectly calculated. Credit alternative coding which produces same results. Accept use of exponent operators: C# e.g. Math.Pow (2, 3) = 8 Java e.g. Math.pow (2, 3) = 8 Python e.g. 2**3 = 8.	(6)

Question	Designated marks:	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 5 of 17 · find part b in this mark scheme extract

Java Example code

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 6 of 17 · find part b in this mark scheme extract

3(a)

```
package q03aFINISHED;

import java.io.BufferedReader;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.PrintWriter;
import java.io.UnsupportedEncodingException;
import java.util.Scanner;

public class Q03aFINISHED
{
    public static void main(String[] args) throws FileNotFoundException,UnsupportedEncodingException
    {
        String email="";
        PrintWriter errorFile;

        //  Open the input file

        Scanner emailFile = new Scanner(new BufferedReader(new FileReader("Email.txt")));

        //  Open the output file

        errorFile = new PrintWriter("Error.txt", "UTF-8");

        //  Find errors and write to output file

        while (emailFile.hasNextLine())
        {
            email = emailFile.nextLine();
            if (!email.contains("@"))
            {
                errorFile.println(email);
            }
        }

        //  Close files

        emailFile.close();
        errorFile.close();
    }
}
```

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 7 of 17 · find part b in this mark scheme extract

3(c)	Note: Line 19 is an alternative response to line 18
	
	`package q03cFINISHED;`
	
	`import java.util.Scanner;`
	
	`public class Q03c`
	`{`
	`  public static void main(String[] args)`
	`  {`
	`  // Write your code here`
	`  Scanner input = new Scanner(System.in);`
	`  int myValue = 1;`
	`  while (myValue >= 1 && myValue <= 50)`
	`  {`
	`  System.out.println("Enter a value from 1 to 50: ");`
	`  myValue = input.nextInt();`
	`  if (myValue >= 1 && myValue <= 50)`
	`  {`
	`  System.out.println("Value " + myValue + " Square " + myValue * myValue + " Cube " + myValue * myValue * myValue);`
	`  System.out.println("Value " + myValue + " Square " + Math.pow(myValue, 2) + " Cube " + Math.pow(myValue, 3));`
	`  }`
	`  }`
	`  }`
	`}`	

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 8 of 17 · find part b in this mark scheme extract

5

```
        int total = 0;
        double average;
        int books[]  = new int[libraryRecord.length];
        int gold = 0, silver = 0, bronze = 0;
        int goldID = 0, silverID = 0, bronzeID = 0;

        System.out.println("Pupil ID of pupils who have read less than 10 books");

            for (int i = 0; i < libraryRecord.length; i++)
            {
                books[i] = Integer.parseInt(libraryRecord[i][3]);
                if (books[i] > gold)
                {
                    bronze = silver; silver = gold; gold = books[i]; goldID = i;
                }
                else if (books[i] > silver)
                {
                    bronze = silver; silver = books[i]; silverID = i;
                }
                else if (books[i] > bronze)
                {
                    bronze = books[i]; bronzeID = i;
                }
                total = total + books[i];

                if (books[i] < 10)
                {
                    System.out.println(libraryRecord[i][0]);
                }
            }
            System.out.println();

            System.out.println("Total books read by pupils = " + total);

            System.out.println();

            average = total / libraryRecord.length;

            System.out.println("Average books read by pupils = " + average);

            System.out.println();

            System.out.println("Gold winner is " + libraryRecord[goldID][1] + " " + libraryRecord[goldID][2]);
            System.out.println("Silver winner is " + libraryRecord[silverID][1] + " " + libraryRecord[silverID][2]);
            System.out.println("Bronze winner is " + libraryRecord[bronzeID][1] + " " + libraryRecord[bronzeID][2]);
```

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 9 of 17 · find part b in this mark scheme extract

Python Example code

1(c)

```
# Q01c

count = 0

maxValue = 5
constantValue = 7

while (count < maxValue):
    print(count, count + constantValue)
    count = count + 1

```

1(f)

```
    if (shopIncome > 5000) or (assistantSales >= 10):
        print ("Assistant bonus = ", assistantSalary * 0.1)

    elif (shopIncome >= 2000) and (assistantSales >= 5):
        print("Assistant bonus = ", assistantSalary * 0.05)
```

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 10 of 17 · find part b in this mark scheme extract

2(a)

```
# Initialise variables
counter = 1
answer = randint(1,10)
guess = 0

# Print prompt and take guess from user
guess = int(input("Enter a number from 1 to 10: "))

# Create while loop to check answer
while guess!=answer:
    counter = counter + 1
    if(guess > answer):
        print (guess, " was too high. Try again.")
    else:
        print (guess, " was too low. Try again.")

    guess = int(input("Guess again: "))

# Report the correct answer to the user and indicate the number of guesses
print ("You guessed", guess, "in", counter, "guesses")
```

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 11 of 17 · find part b in this mark scheme extract

3(a)

```
Q03aFINISHED

#   Open file and input data
with open('Email.txt','r') as inputFile:
    emailList = inputFile.readlines()

#   Open file for output
outputFile = open('Error.txt', 'a')

#   Find errors and write to error log
for address in emailList:
    if not "@" in address:
        outputFile.write(address)

#   Close files
outputFile.close()
inputFile.close()
```

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 12 of 17 · find part b in this mark scheme extract

3(c)

```
1  # Q03c
2
3  # Write your code below this line
4  myValue = 1
5
6  while (myValue >= 1 and myValue <= 50):
7      myValue = int(input("Enter a value from 1 to 50: "))
8      if (myValue >= 1 and myValue <= 50):
9          print("Value: ", myValue, "Square: ",myValue*myValue, "Cube: ", myValue*myValue*
           myValue)
10
11         print("Value: ", myValue, "Square: ", myValue**2, "Cube: ", myValue**3)
```

Note: Line 11 is an alternative response to line 9.

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 13 of 17 · find part b in this mark scheme extract

5

```
# Write your code below this line

total = 0
gold = 0
silver = 0
bronze = 0
goldID = 0
silverID = 0
bronzeID = 0

print ("Pupil ID of pupils who have read less than 10 books ")

for row in range(len(libraryRecord)):
    books = int(libraryRecord[row][3])
    if (books > gold):
        bronze = silver
        silver = gold
        gold = books
        goldID = row
    elif (books > silver):
        bronze = silver
        silver = books
        silverID = row
    elif (books > bronze):
        bronze = books
        bronzeID = row

    total = total + books

    if (books < 10):
        print ("\t",libraryRecord[row][0])

print ("\n")
print ("Total number of books read by pupils: " , total)

print ("\n")
print ("Average number of books read by pupils: " , total / len(libraryRecord[0]))

print ("\n")
print ("Gold medal winner is: \t\t", libraryRecord[goldID][1], " ", libraryRecord[goldID][2])
print ("Silver medal winner is: \t", libraryRecord[silverID][1], " ", libraryRecord[silverID][2])
print ("Bronze medal winner is: \t", libraryRecord[bronzeID][1], " ", libraryRecord[bronzeID][2])
```

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 14 of 17 · find part b in this mark scheme extract

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 15 of 17 · find part b in this mark scheme extract

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 16 of 17 · find part b in this mark scheme extract

Type MS · Compare sorts and searches

Answer 2 of 8 · Question 3 part b · 2 marks

2019 JUN · Paper 02 · extract 17 of 17 · find part b in this mark scheme extract

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Type MS · Compare sorts and searches

Answer 3 of 8 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part b.i in this mark scheme extract

5 (a)	A1 A2 A3 A4	2.1.4	Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g. First 3 characters are upper case letters: Fff123456o Characters 4 to 9 are non-zero numbers: FFF101111o Consists of only 10 characters: FFF111111ee Final character correct for sum of numbers: FFF112233o		(4)

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 3 of 8 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part b.i in this mark scheme extract

5 (b) (i)	B1 B2	1.1.9	Award 1 mark each for any of: • Simple implementation (1) • Can be used for sorted or unsorted lists (1) • If the target is at the beginning of the data structure the search will be faster than a binary search	Accept disadvantages of a binary compared with a linear search.	(2)

Type MS · Compare sorts and searches

Answer 3 of 8 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part b.i in this mark scheme extract

5 (b) (ii)	B3 B4	1.1.9	Award 1 mark each for any of: • May need to compare with all items in list before search complete (1) • May need longer time for searching a large list (1)	Accept advantages of a binary compared with a linear search.	(2)

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Type MS · Compare sorts and searches

Answer 3 of 8 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part b.i in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 3 of 8 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part b.i in this mark scheme extract

5 (c) (i)	C1 C2	2.6.2	Award 1 mark each up to a maximum of 2 for: • Both can use arguments / parameter passing (1) • Both can use local variables (1) • Both can be called from anywhere within the program (1) • Code can be reused without being rewritten (1) • Both can be library files (1) • The code can be independently tested for both (1)		(2)

Type MS · Compare sorts and searches

Answer 3 of 8 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part b.i in this mark scheme extract

5 (c) (ii)	C3 C4	2.6.2	Award two marks for a linked explanation such as: • A function must always return a result (1) whereas a procedure does not (1) • A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1) • The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1) • A function produces information (1) whereas a procedure performs a task (1)		(2)

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Type MS · Compare sorts and searches

Answer 3 of 8 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part b.i in this mark scheme extract

Mark Scheme – Python Coding

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part b.ii in this mark scheme extract

5 (a)	A1 A2 A3 A4	2.1.4	Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g. First 3 characters are upper case letters: Fff123456o Characters 4 to 9 are non-zero numbers: FFF101111o Consists of only 10 characters: FFF111111ee Final character correct for sum of numbers: FFF112233o		(4)

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part b.ii in this mark scheme extract

5 (b) (i)	B1 B2	1.1.9	Award 1 mark each for any of: • Simple implementation (1) • Can be used for sorted or unsorted lists (1) • If the target is at the beginning of the data structure the search will be faster than a binary search	Accept disadvantages of a binary compared with a linear search.	(2)

Type MS · Compare sorts and searches

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part b.ii in this mark scheme extract

5 (b) (ii)	B3 B4	1.1.9	Award 1 mark each for any of: • May need to compare with all items in list before search complete (1) • May need longer time for searching a large list (1)	Accept advantages of a binary compared with a linear search.	(2)

Type MS · Compare sorts and searches

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part b.ii in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part b.ii in this mark scheme extract

5 (c) (i)	C1 C2	2.6.2	Award 1 mark each up to a maximum of 2 for: • Both can use arguments / parameter passing (1) • Both can use local variables (1) • Both can be called from anywhere within the program (1) • Code can be reused without being rewritten (1) • Both can be library files (1) • The code can be independently tested for both (1)		(2)

Type MS · Compare sorts and searches

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part b.ii in this mark scheme extract

5 (c) (ii)	C3 C4	2.6.2	Award two marks for a linked explanation such as: • A function must always return a result (1) whereas a procedure does not (1) • A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1) • The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1) • A function produces information (1) whereas a procedure performs a task (1)		(2)

8 of 16

Type MS · Compare sorts and searches

Answer 4 of 8 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part b.ii in this mark scheme extract

Mark Scheme – Python Coding

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 5 of 8 · Question 4 part b.i · 2 marks

2021 JUN · Paper 02 · extract 1 of 7 · find part b.i in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
4 (b)(i)	B1	Award up to 2 marks for a linked explanation: • binary search can be quicker / is more effective than a linear search / is more effective with larger lists (1) as it does not have to examine each item in the list (1) • binary search halves the list each time (1) so it can be faster to find an item (1) • binary search requires fewer comparisons than a linear search to establish an item is not in the list (1) because the linear list would need to compare each item before establishing this (1)		(2)

Type MS · Compare sorts and searches

Answer 5 of 8 · Question 4 part b.i · 2 marks

2021 JUN · Paper 02 · extract 2 of 7 · find part b.i in this mark scheme extract

would need to compare each item before establishing this (1) (2)

4 (b)(ii)

Correct answer

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	1
5	mil19	
6	rib10	2
7	str15	3
8	tor16	

Award one mark for each correct value in order column (4)

B1	Start of search correct	Accept 5 and 7 for B1 and B2 (2 marks)
B2	Second search item correct	Follow through if start of search incorrect
B3	Third search item correct	Follow through if start of search incorrect
B4	All correct	

Type MS · Compare sorts and searches

Answer 5 of 8 · Question 4 part b.i · 2 marks

2021 JUN · Paper 02 · extract 3 of 7 · find part b.i in this mark scheme extract

		All correct		
4 (b)(iii)	B5	Award 1 mark for: $2 \text{ or } \log_2 n + 1$		(1)

Type MS · Compare sorts and searches

Answer 5 of 8 · Question 4 part b.i · 2 marks

2021 JUN · Paper 02 · extract 4 of 7 · find part b.i in this mark scheme extract

		or $\log_2 n + 1$		(1)
4 (b)(iv)	B6	Award 1 mark for any of: • bubble sort • merge sort	Accept any known sorting algorithm	(1)

Type MS · Compare sorts and searches

Answer 5 of 8 · Question 4 part b.i · 2 marks

2021 JUN · Paper 02 · extract 5 of 7 · find part b.i in this mark scheme extract

Python Code Mark Scheme

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 5 of 8 · Question 4 part b.i · 2 marks

2021 JUN · Paper 02 · extract 6 of 7 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	A1	At least 1 variable has a meaningful name	Ignore spelling mistakes in input message	(7)
	A2	Product name requested using a suitable input message		
	A3	Random number generated that would be at least 10 **or** no higher than 30		
	A4	Random number generated that would be in the correct range 10 to 30		
	A5	First 3 letters of product name generated		
	A6	First 3 letters of product name and random number concatenated to generate productCode		
	A7	productCode and productName output in the same print statement		
Code examples				
Python				

```
# Get input
productName = input("Enter the product name: ")

randomNum = 0

# Generate a random number between 10 and 30 inclusive
randomNum = random.randint(10,30)

# Generate the product code - first three letters of p
productCode = productName[0:3] + str(randomNum)

# Display the product code and the product name
print(productCode + " " + productName)
```

Type MS · Compare sorts and searches

Answer 5 of 8 · Question 4 part b.i · 2 marks

2021 JUN · Paper 02 · extract 7 of 7 · find part b.i in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	ran	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 6 of 8 · Question 4 part b.iv · 1 mark

2021 JUN · Paper 02 · extract 1 of 7 · find part b.iv in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
4 (b)(i)	B1	Award up to 2 marks for a linked explanation: • binary search can be quicker / is more effective than a linear search / is more effective with larger lists (1) as it does not have to examine each item in the list (1) • binary search halves the list each time (1) so it can be faster to find an item (1) • binary search requires fewer comparisons than a linear search to establish an item is not in the list (1) because the linear list would need to compare each item before establishing this (1)		(2)

Type MS · Compare sorts and searches

Answer 6 of 8 · Question 4 part b.iv · 1 mark

2021 JUN · Paper 02 · extract 2 of 7 · find part b.iv in this mark scheme extract

would need to compare each item before establishing this (1) (2)

4 (b)(ii)

Correct answer

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	1
5	mil19	
6	rib10	2
7	str15	3
8	tor16	

Award one mark for each correct value in order column (4)

B1	Start of search correct	Accept 5 and 7 for B1 and B2 (2 marks)
B2	Second search item correct	Follow through if start of search incorrect
B3	Third search item correct	Follow through if start of search incorrect
B4	All correct	

Type MS · Compare sorts and searches

Answer 6 of 8 · Question 4 part b.iv · 1 mark

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		All correct		
4 (b)(iii)	B5	Award 1 mark for: $3 \text{ or } \log_2 n + 1$		(1)

Type MS · Compare sorts and searches

Answer 6 of 8 · Question 4 part b.iv · 1 mark

2021 JUN · Paper 02 · extract 4 of 7 · find part b.iv in this mark scheme extract

		or $\log_2 n + 1$		(1)
4 (b)(iv)	B6	Award 1 mark for any of: • bubble sort • merge sort	Accept any known sorting algorithm	(1)

Type MS · Compare sorts and searches

Answer 6 of 8 · Question 4 part b.iv · 1 mark

2021 JUN · Paper 02 · extract 5 of 7 · find part b.iv in this mark scheme extract

Python Code Mark Scheme

Type MS · Compare sorts and searches

Answer 6 of 8 · Question 4 part b.iv · 1 mark

2021 JUN · Paper 02 · extract 6 of 7 · find part b.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	A1	At least 1 variable has a meaningful name	Ignore spelling mistakes in input message	(7)
	A2	Product name requested using a suitable input message		
	A3	Random number generated that would be at least 10 **or** no higher than 30		
	A4	Random number generated that would be in the correct range 10 to 30		
	A5	First 3 letters of product name generated		
	A6	First 3 letters of product name and random number concatenated to generate productCode		
	A7	productCode and productName output in the same print statement		
Code examples				
Python				

```
# Get input
productName = input("Enter the product name: ")

randomNum = 0

# Generate a random number between 10 and 30 inclusive
randomNum = random.randint(10,30)

# Generate the product code - first three letters of p
productCode = productName[0:3] + str(randomNum)

# Display the product code and the product name
print(productCode + " " + productName)
```

Type MS · Compare sorts and searches

Answer 6 of 8 · Question 4 part b.iv · 1 mark

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For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question		Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 1 of 9 · find part b.ii in this mark scheme extract

5 (b)(i)	B1 B2 B3	**Award one mark for each:** • Pass 1 correct in first row of the response (1) • Pass 2 **OR** 3 correct in any row of the response (1) • All passes correct and no extra passes (1) 2 3 8 1 9 10 2 3 1 8 9 10 2 1 3 8 9 10 1 2 3 8 9 10	If the final row (1, 2, 3, 8, 9, 10) is repeated treat as the same row.	(3)

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 2 of 9 · find part b.ii in this mark scheme extract

5 (b)(ii)	B4 B5	**Award up to 2 marks for a linked explanation such as:** • The sorting is done in the same place as the original data (1), which means only one additional variable is needed to store the value being swapped (when the values are out of order (1) • All sort and swapping operations are done on the original data (1) which means the amount of memory needed is constant (1) • The list does not need to be split (1), which would use more memory (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 3 of 9 · find part b.ii in this mark scheme extract

5 (c)	C1 C2 C3	**Award up to 3 marks for a linked description that includes:** • Start at the beginning of the list (1) • Compare each value in the list with the search item (1) • Repeat until a match is found / the end of the list is reached (1)		(3)

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 4 of 9 · find part b.ii in this mark scheme extract

Code - Python

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 5 of 9 · find part b.ii in this mark scheme extract

5 (a)	**Award 1 mark for each of:**			
	A1	Variable to store a number of incorrect passwords (1)		
	A2	Loop used (1)		
	A3	Each password checked to see if the first character is an uppercase letter (1)		
	A4	Each password checked to see if it contains a digit (1)		
	A5	Only check for digits if first character is uppercase OR Only checks if first character is uppercase if at least one digit is present (1)		
	A6	Number of incorrect passwords incremented correctly (1) (allow follow through)		
	A7	All incorrect passwords displayed (1) (allow follow through but must try to display those without uppercase and those without digits)		
	A8	Text file closed (1)		
	A9	Number of incorrect passwords displayed (1) (allow follow through)		(9)
Code examples				
Python				

```
# Add your code here
incorrectPasswords = 0

for password in passwordFile:
    valid = False

    if password[0] in alphabet:
        index = 1
        while (valid == False) and (index in range (1, len(password))):
            if password[index]in digit:
                valid = True
            else:
                index = index + 1

    if not valid:
        incorrectPasswords = incorrectPasswords + 1
        print(password)

print(incorrectPasswords,"passwords are incorrect.")

passwordFile.close()
```

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 6 of 9 · find part b.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 7 of 9 · find part b.ii in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
            # If the letter is in the secret word and not already in the correctletters, add it
            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongLetters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 8 of 9 · find part b.ii in this mark scheme extract

Main program

```
# Add your main program code here
wordLength = len(wordToGuess)
maxAttempts = wordLength + 3
numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

Type MS · Compare sorts and searches

Answer 7 of 8 · Question 5 part b.ii · 2 marks

2022 JUN · Paper 02 · extract 9 of 9 · find part b.ii in this mark scheme extract

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Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 1 of 9 · find part c in this mark scheme extract

5 (b)(i)	B1 B2 B3	**Award one mark for each:** • Pass 1 correct in first row of the response (1) • Pass 2 **OR** 3 correct in any row of the response (1) • All passes correct and no extra passes (1) 2 3 8 1 9 10 2 3 1 8 9 10 2 1 3 8 9 10 1 2 3 8 9 10	If the final row (1, 2, 3, 8, 9, 10) is repeated treat as the same row.	(3)

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 2 of 9 · find part c in this mark scheme extract

5 (b)(ii)	B4 B5	**Award up to 2 marks for a linked explanation such as:** • The sorting is done in the same place as the original data (1), which means only one additional variable is needed to store the value being swapped (when the values are out of order (1) • All sort and swapping operations are done on the original data (1) which means the amount of memory needed is constant (1) • The list does not need to be split (1), which would use more memory (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 3 of 9 · find part c in this mark scheme extract

5 (c)	C1 C2 C3	**Award up to 3 marks for a linked description that includes:** • Start at the beginning of the list (1) • Compare each value in the list with the search item (1) • Repeat until a match is found / the end of the list is reached (1)		(3)

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 4 of 9 · find part c in this mark scheme extract

Code - Python

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 5 of 9 · find part c in this mark scheme extract

5 (a)	**Award 1 mark for each of:**			
	A1	Variable to store a number of incorrect passwords (1)		
	A2	Loop used (1)		
	A3	Each password checked to see if the first character is an uppercase letter (1)		
	A4	Each password checked to see if it contains a digit (1)		
	A5	Only check for digits if first character is uppercase OR Only checks if first character is uppercase if at least one digit is present (1)		
	A6	Number of incorrect passwords incremented correctly (1) (allow follow through)		
	A7	All incorrect passwords displayed (1) (allow follow through but must try to display those without uppercase and those without digits)		
	A8	Text file closed (1)		
	A9	Number of incorrect passwords displayed (1) (allow follow through)		(9)
Code examples				
Python				

```
# Add your code here
incorrectPasswords = 0

for password in passwordFile:
    valid = False

    if password[0] in alphabet:
        index = 1
        while (valid == False) and (index in range (1, len(password))):
            if password[index]in digit:
                valid = True
            else:
                index = index + 1

    if not valid:
        incorrectPasswords = incorrectPasswords + 1
        print(password)

print(incorrectPasswords,"passwords are incorrect.")

passwordFile.close()
```

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 6 of 9 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 7 of 9 · find part c in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
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            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongletters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 8 of 9 · find part c in this mark scheme extract

Main program

```
# Add your main program code here
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numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

Type MS · Compare sorts and searches

Answer 8 of 8 · Question 5 part c · 3 marks

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